



32-bit microcontroller

XH-Link-Writer programmer

User manual

Rev1.1 November 2023

Applicable objects

Product series	Product model	Product series	Product model
HC32F460	HC32F460PETB HC32F460PEHB HC32F460KETA HC32F460KEUA HC32F460JETA HC32F460JEUA HC32F460PCTB HC32F460KCTA HC32F460JCTA	HC32F4A0	HC32F4A0TIHB HC32F4A0SITB HC32F4A0SIHB HC32F4A0RITB HC32F4A0PITB HC32F4A0PIHB HC32F4A0SGT8 HC32F4A0SGHB HC32F4A0RGTB HC32F4A0PGTB
HC32L110	HC32L110C6PA HC32L110C6UA HC32L110B6PA HC32L110B6YA HC32L110C4PA HC32L110C4UA HC32L110B4PA	HC32L130/136	HC32L136K8TA HC32L136J8TA HC32L130J8TA HC32L130J8UA HC32L130F8UA HC32L130E8PA
HC32L072/073	HC32L072PATA HC32L072KATA HC32L072JATA HC32L072FAUA HC32L073PATA HC32L073KATA HC32L073JATA	-	-

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1 Introduction

XH-Link-Writer is an online programming environment for the Cortex-M series MCU of Xiaohua Semiconductor (XHSC). It supports all Cortex-M series MCU products of Xiaohua Semiconductor, and aims to provide users with a compact, portable, safe, reliable and easy-to-operate programming tool.

The XH-Link-Writer system is shown in Figure 1-1. The XH-Link-Writer software programs the target MCU through the XH-LINK motherboard.

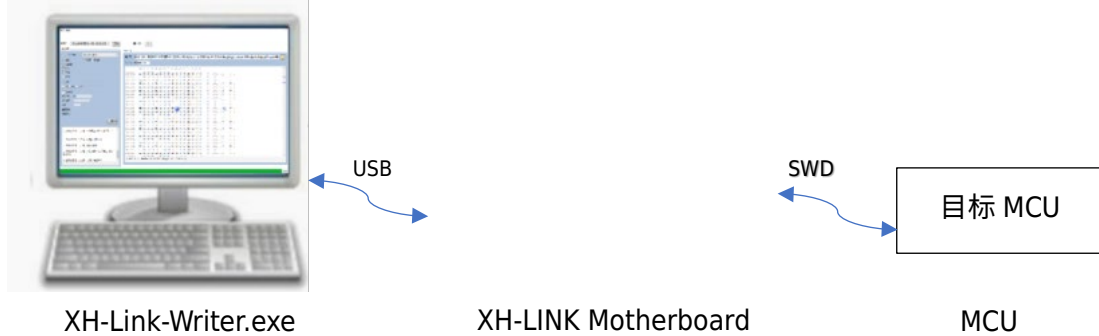


Figure 1-1 XH-Link-Writer System Connection Diagram

1.1 XH-Link-Writer Features

The functions supported by XH-Link-Writer are as follows:

- 1) Chip programming parameters can be configured:
 - Chip model selection
 - Programming action
 - Chip erase/page erase
 - Read protection
 - Rolling code function
 - Reset function
 - Programming speed (1M/ 5M/ 10M)
- 2) Chip programming parameter configuration can be saved for quick use next time startup.
- 3) The programming interface supports SWD.

1.2 XH-LINK Features

For the functions and features supported by XH-LINK, please refer to the document "XH-LINK Debugger User Manual RevX.X.pdf".

2 XH-Link-Writer Description

2.1 Programming Method

XH-Link-Writer currently only supports SWD programming mode. For SWD connection methods, please refer to the document "XHLINK Debugger User Manual RevX.X.pdf".

2.2 Software Interface

Double-click "XH-Link-Writer.exe" to open the software. The software interface is shown in Figure 2-1.

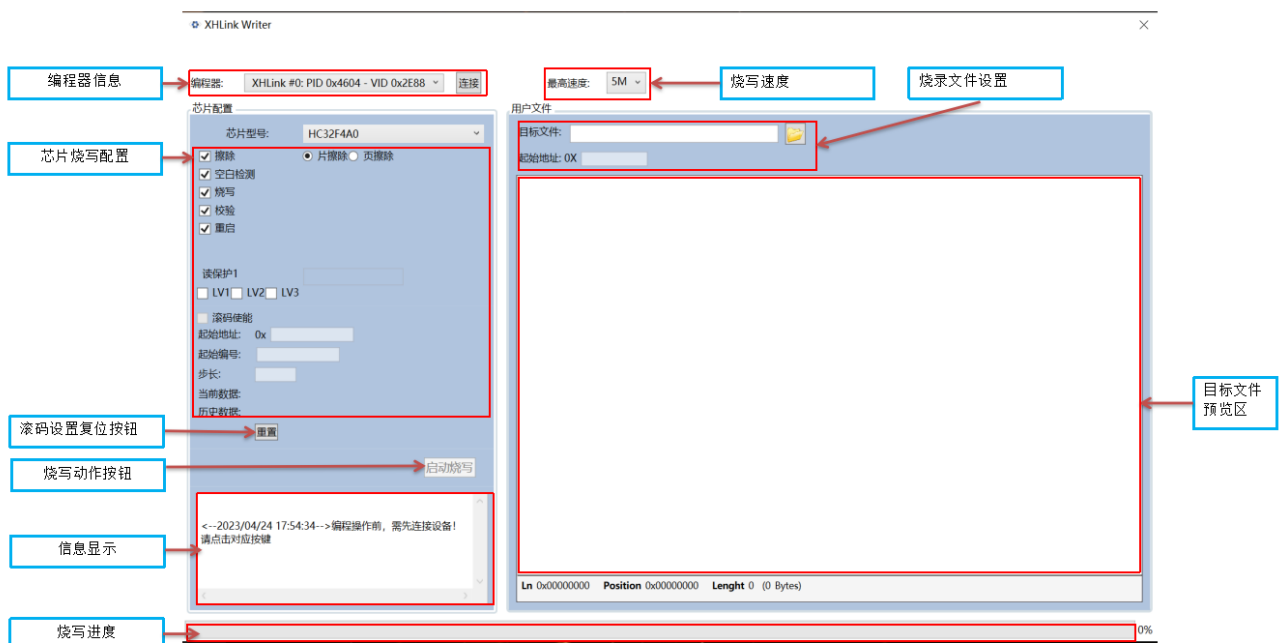


Figure 2-1 Software interface

XH-Link-Writer functions and features are as follows:

- Automatically identify the programmer and display it in the list. You need to click the "Connect" button before programming.
- The target MCU model can be set in the chip configuration area. After manually selecting the MCU model, the target file selection/read protection setting/rolling code enable setting need to be reconfigured. The MCU model can be saved to the configuration file.
- Erase: Erasing supports two methods: chip erase and page erase. When chip erase is selected, the full chip erase method is used; when page erase is selected, the corresponding area is erased according to the size of the programmed file.
- Blank detection: Check whether the entire chip flash is in a blank state (all 0xFF) to ensure that the flash erase is valid. It is enabled by default.
- Program: The operation of programming the target file to the MCU flash is enabled by default.
- Verification: Verify whether the data programmed to the flash is consistent with the programmed target file. It is enabled by default.
- Restart: After the firmware is successfully programmed, the MCU will be reset and run automatically. The modified settings can be saved.

- Read protection 1: Read protection is implemented for the flash area to prevent untrusted users from reading flash data. After protection, flash data can be read with a password. Different MCU models support different protection levels. The protection level can be saved to the configuration file. The password will not be saved. The password needs to be entered again when the software is restarted.
- After the rolling code function is enabled, the rolling code parameters need to be set. The rolling code parameters include the rolling code starting address, starting number, and step length. The enable status/historical burning number and step length can be saved to the configuration file.
- Rolling code starting address: The rolling code number is written to the address of the MCU flash, occupying 4 bytes of space. The address must be within the valid range of the MCU flash and does not overlap with the firmware itself. The starting address can be saved to the configuration file.
- Rolling code starting number: The size occupies 4 bytes of space, the format is decimal input, the valid range is 0 to 4294967295 (hexadecimal 0xFFFF FFFF), and it will automatically return to zero after reaching the maximum value. The current data burned can be saved to the configuration file.
- Rolling code step: The step value between each number. The set data can be saved to the configuration file. The effective range of the step is 0 to 999
- Rolling code current data: Displays the number written in this programming. This data can be saved to the configuration file as historical data.
- Rolling code historical data: Displays the number successfully written last time.
- Reset button: When the user does not need to use the saved rolling code configuration information setting, the rolling code function will no longer be enabled after reset. It can be re-enabled when needed.
- The information prompt area displays operation prompts and operation progress information.
- The programming speed is the maximum frequency of the SWD clock. There are three settings to choose from: 1M/ 5M/ 10M. The higher the speed, the shorter the theoretical programming time.
- The target file area is the firmware file path to be programmed to the MCU system. You can choose files in hex and bin formats. If you choose bin format, you need to set a legal and valid starting address. It is recommended to use hex format. XH-Link-Writer automatically recognizes the firmware address.
- Click the Start Programming button to start the programming action.
- The programming progress area at the bottom displays the programming progress in real time, with a percentage format prompt.

3 Driver Installation

This software requires the minimum version of Microsoft.NET Framework v4.7.2. If the software has corresponding problems, you need to download and install a version above V4.7.2.

If the operating system is not installed, please go to the Microsoft official website to download and select the corresponding version to download, as follows:



Figure 3-1 Framework 4.7.2

Download the XH-Link-Writer software. The folder directory is as shown in Figure 3-2:

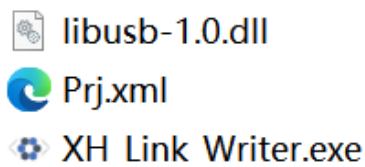


Figure 3-2 XH-Link-Writer file directory

"Prj.xml" user configuration file. Used to save user-related settings.

"XH-Link-Writer.exe" is the execution program file.

"libusb-1.0.dll" is the Windows dll file.

4 Operation Steps

Take the MCU model HC32F460JETA as an example to illustrate the hardware connection and software settings.

4.1 Hardware Connection

Take the EV_F460_LQ100_Rev2.0 EVB board as an example, the XH-Link-Writer programming hardware interface DIO/ CLK/ GND/ RST is connected to the DIO/ CLK/ GND/ RST on the EVB J21 pin header, the EVB uses the XH-LINK 3.3V power supply, and the 3.3V on J21 is connected to the programming hardware interface 3.3V. The actual connection is shown in Figure 4-1.

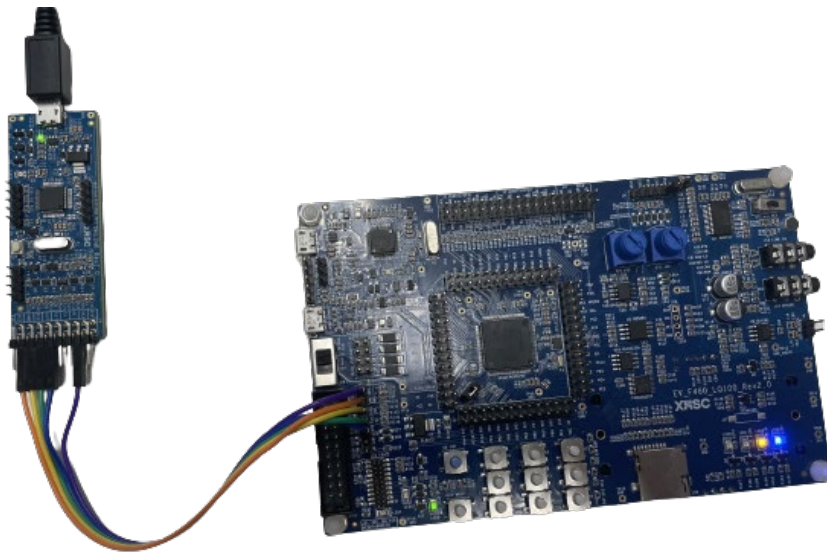


Figure 4-1 XH-LINK/ EV_F460_LQ100_Rev2.0 SWD interface connection

4.2 Open Software

Double-click to open the XH-Link-Writer.exe software and connect. The default maximum speed is 5M. Select the XH-LINK programmer in the programmer list. Click the "Connect" button, and the message "Device opened successfully" will be displayed, as shown in Figure 4-2.

Note: After clicking the "Connect" button successfully, it will turn into "Disconnect". If you click "Connect" continuously and "Disconnect" does not appear, you can manually press the XH-LINK reset button and try to connect again.

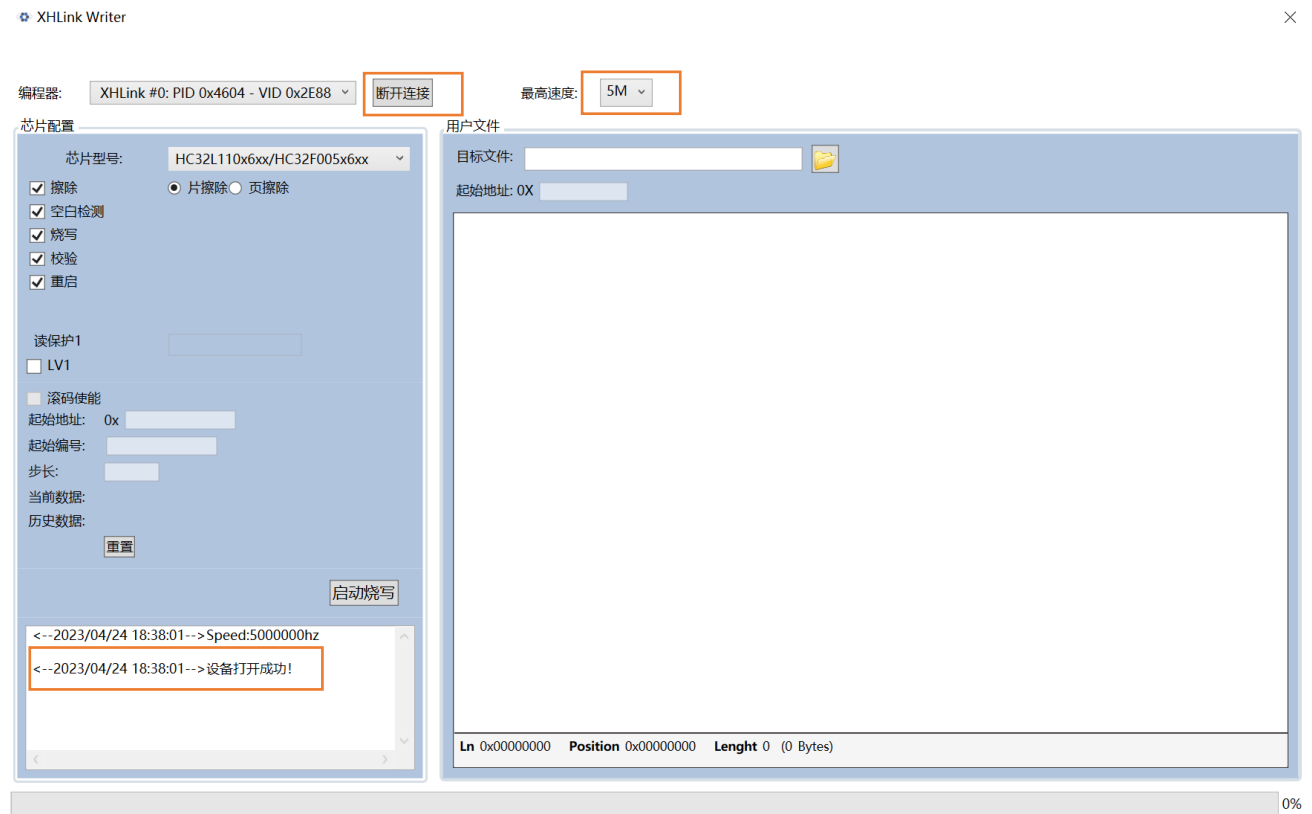


Figure 4-2 Connection successful

4.3 Configure Target MCU Model

Select an MCU model corresponding to the programming target board from the chip model drop-down box, such as HC32F460xExx/HC3245xxExx, as shown in Figure 4-3.

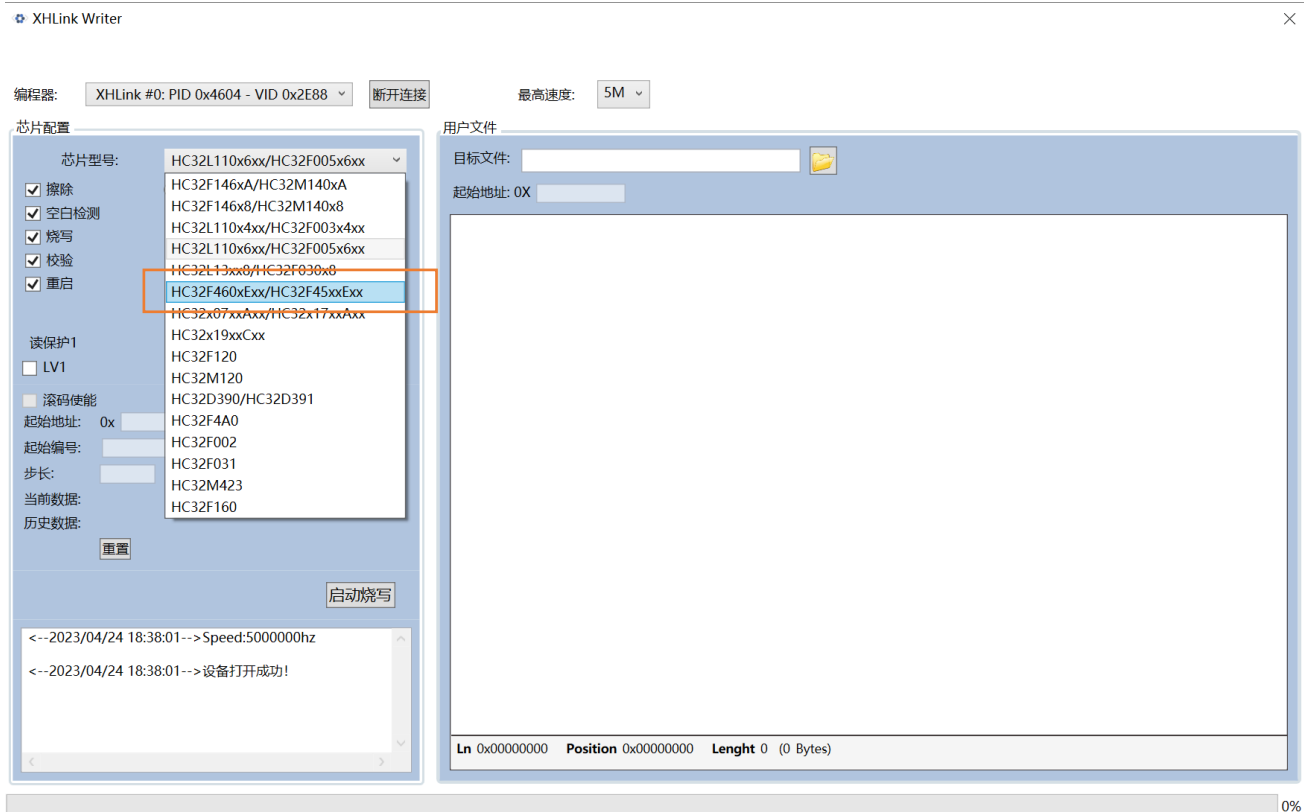


Figure 4-3 MCU model selection

4.4 Configure Programming Process Settings

The erase mode uses the default chip erase, and the blank detection/burning/verification/restart functions are enabled, as shown in Figure 4-4.

Note: After the MCU model HC32F460xExx/ HC32F45xxExx or HC32F4A0 is read-protected, you must select chip erase to program successfully when using XH-Link-Writer again.

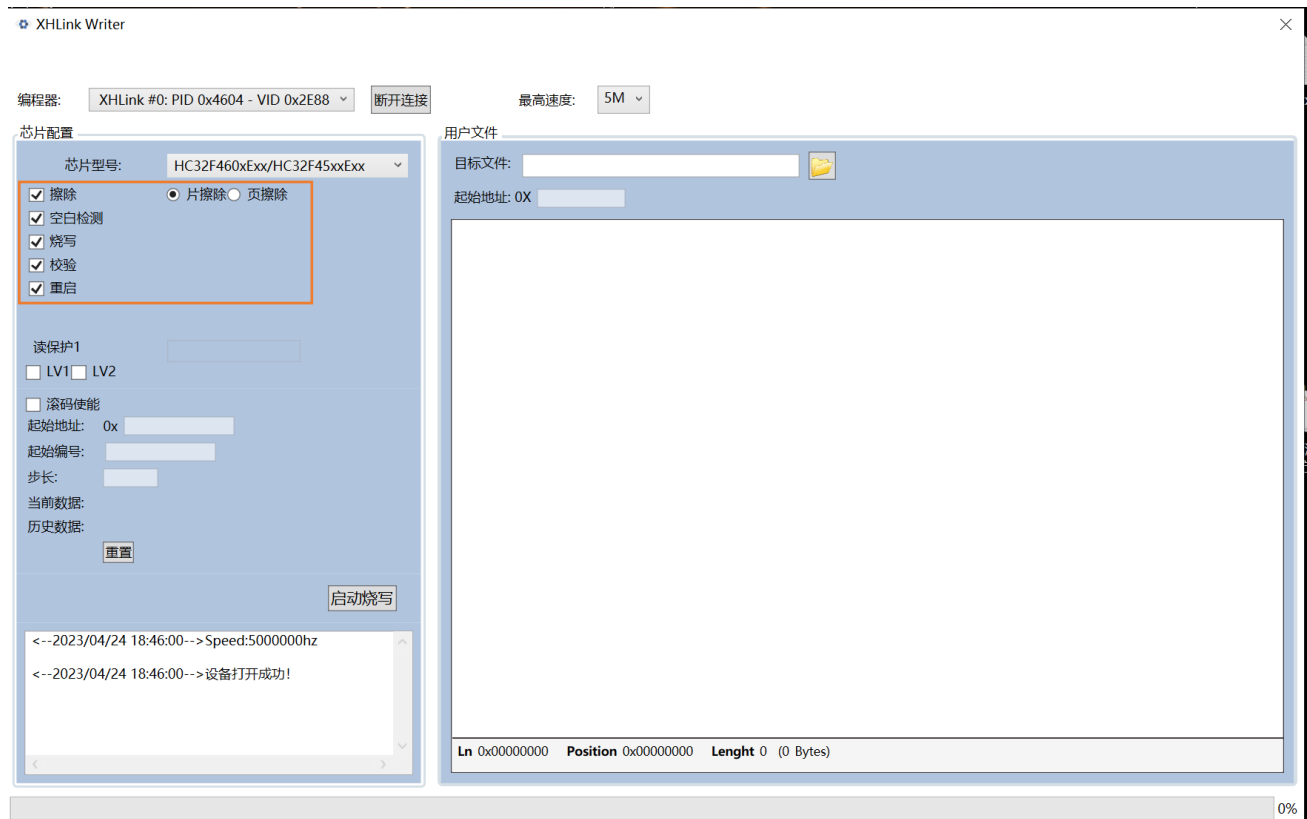


Figure 4-4 Programming process configuration

4.5 Serial Number Function Settings

Check the serial number enable, set the serial number address to 0x1000, the serial number start number to 1, and the step length to 1, as shown in Figure 4-5.

Note: When the serial number start address conflicts with the address within the firmware size range, a message box will be prompted to confirm to prevent overwriting of valid firmware data; when there is historical data, the serial number is automatically programmed with the historical number/step length setting value. When you need to reset it, you can use the following two methods:

- After reselecting the MCU model, check the rolling code enable and set the parameters.
- After clicking the reset button, check the rolling code enable and set the parameters.

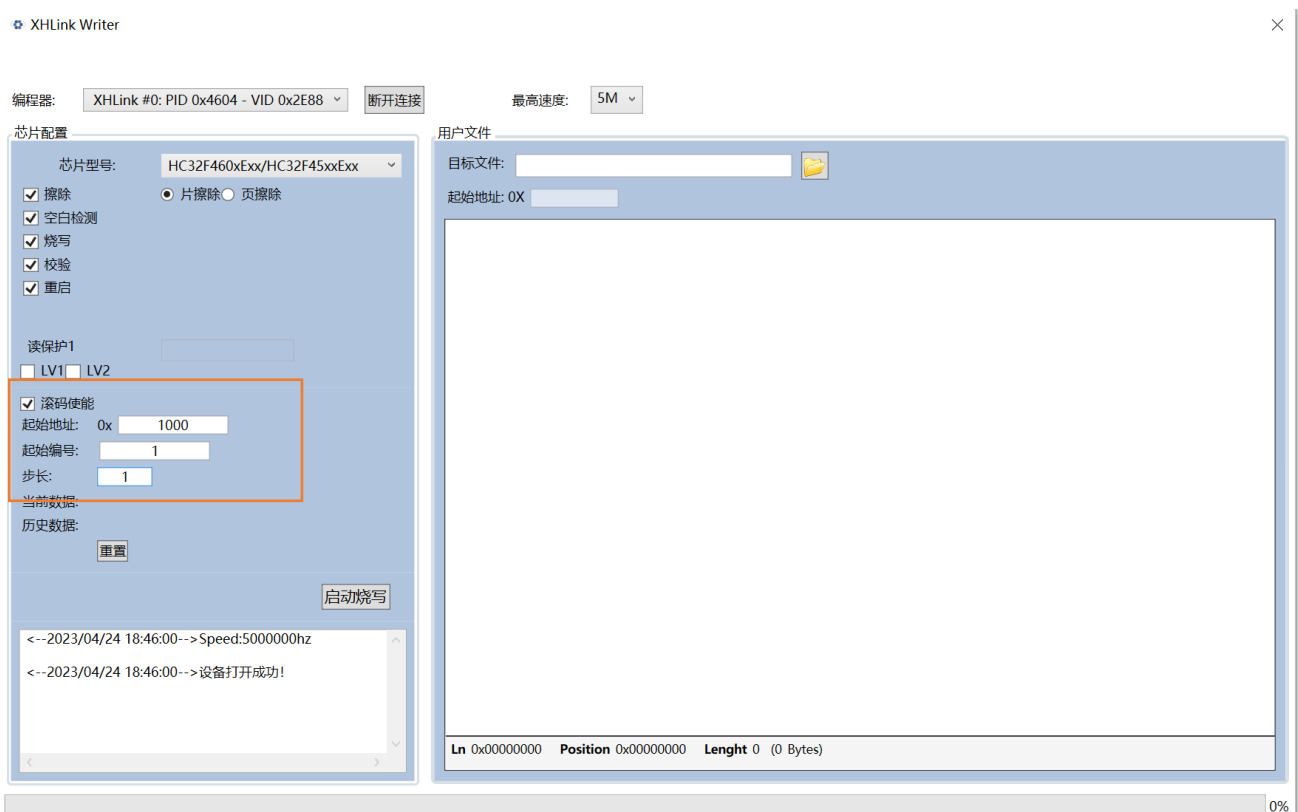


Figure 4-5 Serial number setting

4.6 Set Programming Target File

The target file format supports hex and bin formats. When the target file is in hex format, there is no need to configure the start address, as shown in Figure 4-6. When the target file is in bin format, the start address is configured in hexadecimal by default, as shown in Figure 4-7. After setting, you can preview the firmware data.

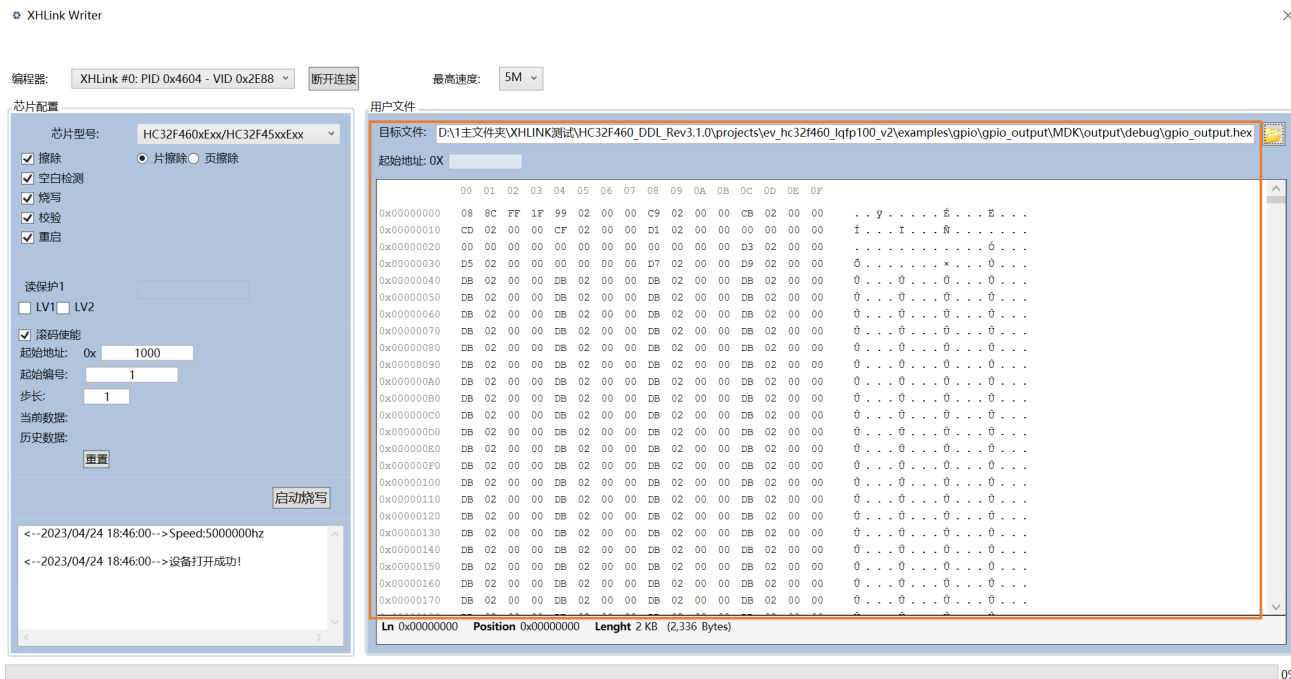


Figure 4-6 Hex format programming file settings

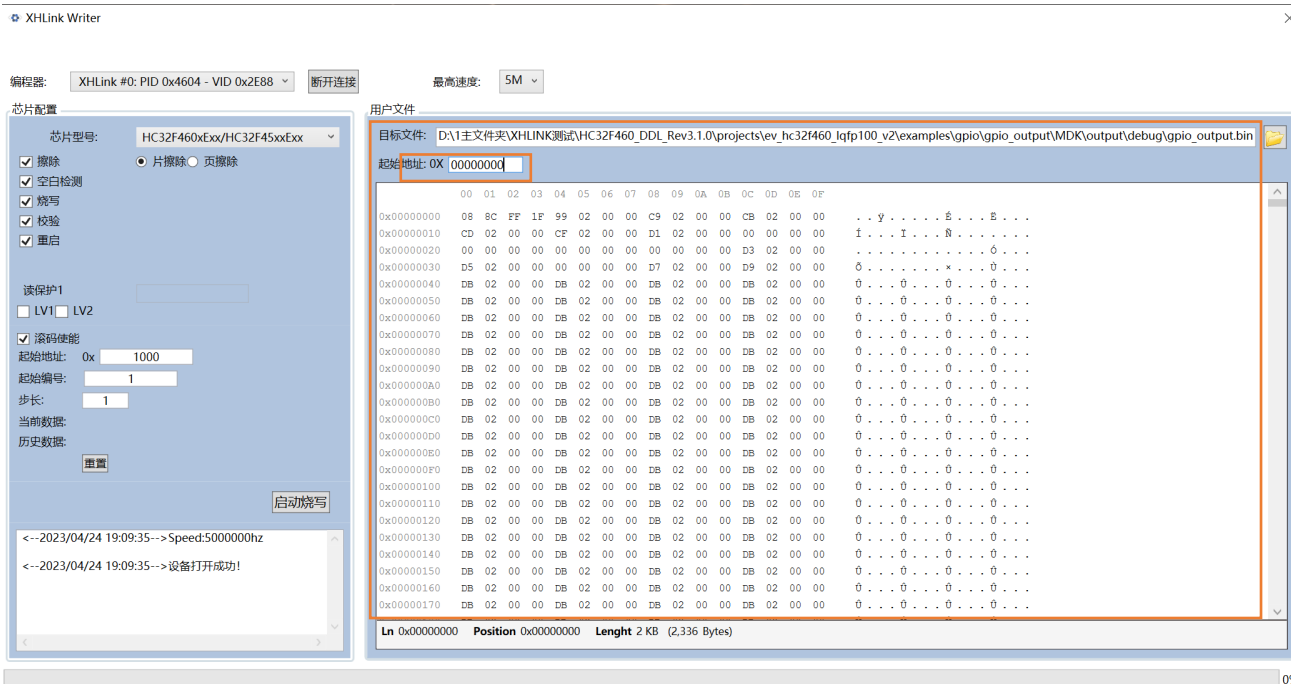


Figure 4-7 Bin format programming file settings

4.7 Read Protection 1 Settings

The supported level of read protection varies according to the MCU model. XH-Link-Writer automatically identifies the supported level, which can be up to Lv3. For Lv1, the password length must be 96 bits, that is, 12 characters, as shown in Figure 4-8.

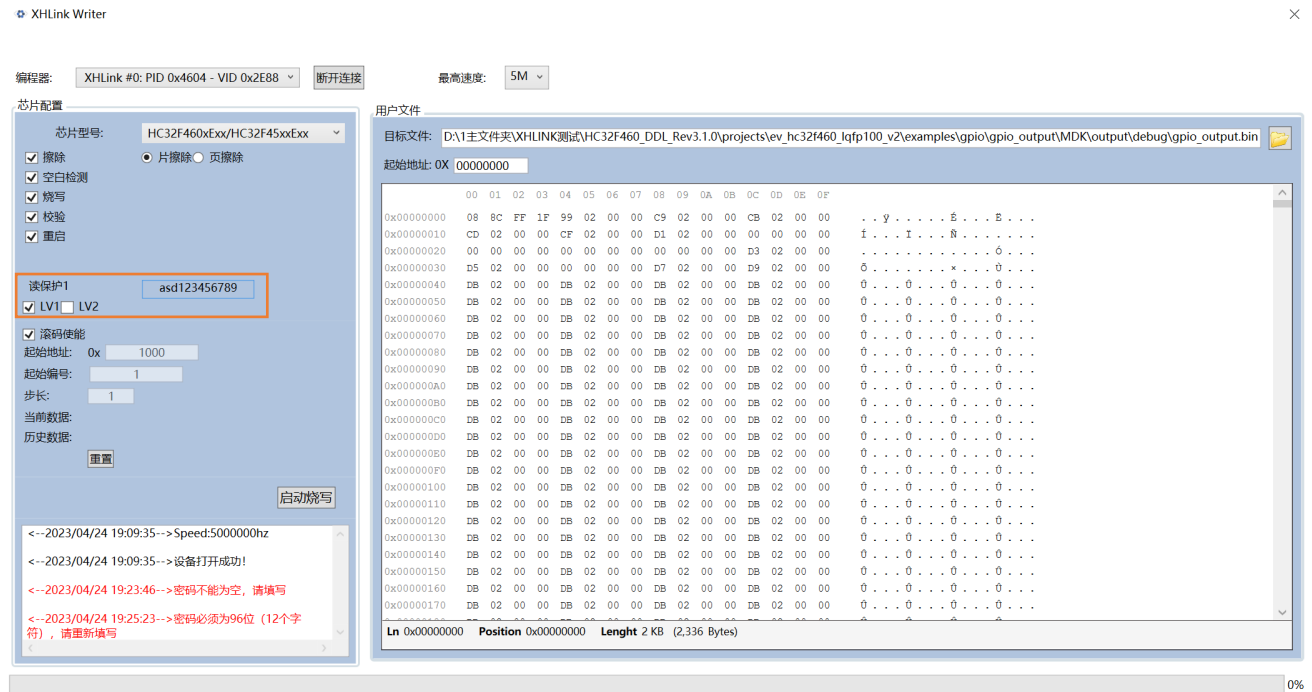


Figure 4-8 Read protection settings

4.8 Start Programming

After clicking the “Start Programming” button, the programming progress bar at the bottom of the XH-Link-Writer displays the programming progress percentage in real time, and the information area displays the programming success information and connection time, as shown in Figure 4-9.

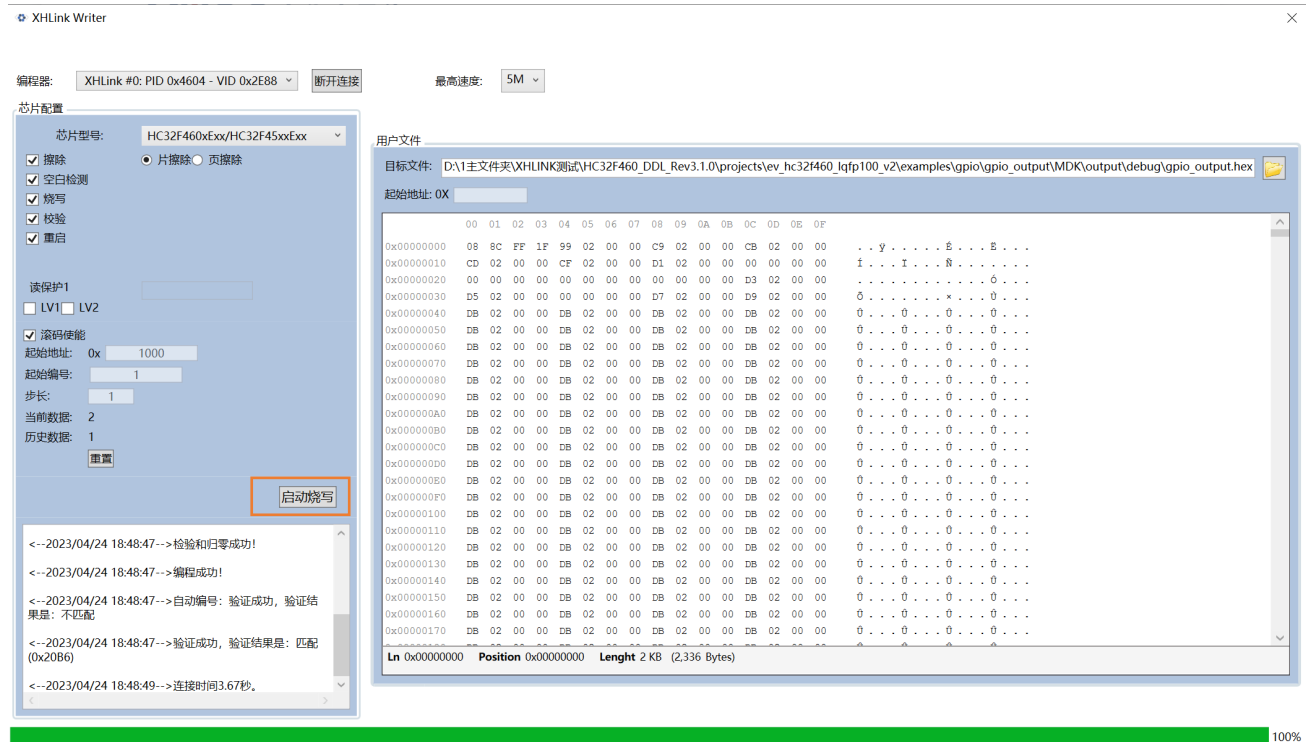


Figure 4-9 Start programming

4.9 Software Exit

You can exit the software by clicking the × button in the upper right corner, and then a prompt window will appear to confirm whether to save the configuration of this burning. After clicking the OK button, the configuration information is saved to the Prj.xml file. The next time you open the XH-Link-Writer software, the last saved configuration information will be automatically loaded; click Cancel to not save, as shown in Figure 4-10.

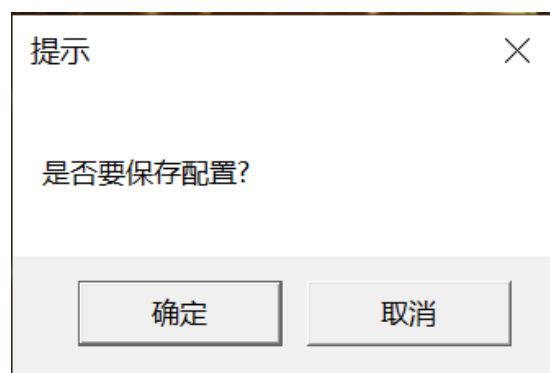


Figure 4-10 Save configuration prompt

5 Error Handling

5.1 Tips

Table 5-1 Tips

Prompt information	Description	Processing
Device opened successfully	Connected to XH-LINK successfully	-
Device closed	Disconnected from XH-LINK successfully	-
Before programming, you need to connect the device first! Please click the corresponding button	After opening the software, please connect the XH-LINK device first.	-
Unspecified Bin File address!	When the target file is selected as a bin format file, you need to manually enter the programming start address	-
Target chip Halt failed!	The chip is in protection state	-
The automatic numbering address overlaps with the user code. Do you want to continue the operation?	The rserial number start address overlaps with the firmware effective address, and you are asked whether to continue automatic numbering.	Click "Yes" to continue automatic numbering, otherwise stop writing numbers
Target chip not found!	The chip is in protection state or the chip is abnormal.	Release the protection or check if the main body is abnormal
Blank check successful! Check result: XX	Flash blank check result	-

5.2 Errors

Table 5-2 Errors

Error message	Description	Resolving
Please select the file you want to burn!	No target file was selected	Select the target file
The file path is incorrect or the file is invalid!	The selected target file path does not exist or the file is invalid or occupied	Reselect the target file and make sure it is valid and not in use.
Hex file format error!	Wrong hex file format	Confirm whether the hex file is correct
Lost connection with device! Please restart!	XH-Link-Writer cannot connect to the programmer	Press the XH-LINK reset button
The password cannot be empty, please fill in	Password field is empty when password is required	Enter the correct password
The parameter cannot be empty, please fill it in! Or the parameter length exceeds 5	When the serial number enable option is checked, the starting address must be filled in correctly	Fill in the legal starting address correctly
The start number parameter cannot be greater than 0xFFFF FFFF, please re-enter	The value of the serial number writing start number setting is out of range	Fill in legal and correct format values
Bin file address is illegal, please fill it in again	The starting address value of the bin file is illegal	Correctly fill in the legal and correct format value
The parameter cannot be empty, please fill it in! Or the parameter length exceeds 3	The step length of the serial number is too long	Set the legal step value
File error! The length exceeds the Flash size of the selected chip.	The size of the burning file exceeds the selected chip Flash size	Reselect the correct burning file or reselect the correct target MCU
Page erase chip failed	Page erase mode erase failed	Check the hardware connection, check the programming process, and try to power on again
Verification was successful,	Verify after firmware is programmed	Check the hardware connection, check the programming process, and try

Error message	Description	Resolving
and the verification result was: No match		to power on again
Verification failed	Verify after firmware is programmed	Check the hardware connection, check the programming process, and try to power on again
Verification and zeroing failed!	Verification and checksum failed	Check the hardware connection, check the programming process, and try to power on again
Programming failed	Programming Failure	Check the hardware connection, check the programming process, and try to power on again

Revision History

Version	Date	Revision Content
Rev1.0	2023/08/07	Initial release.
Rev1.1	2023/11/22	1) Delete the description of the XHLINK function related to hardware. 2) The latest version of .NET Framework is required to modify the software